



OPENAI CODEX HACKATHON 2026 • DEVELOPER TOOLS

Website Doctor

Your website, diagnosed in minutes.

An AI-powered website QC platform. Paste a URL — get a full health diagnosis: real crawling, 60+ automated checks, Lighthouse, axe-core, security probes, and AI-written fixes tied to the exact broken elements.

Your website is losing money right now — and you have no way to know.

Invisible

A missing security header, an unlabeled booking form, a 4 MB hero image — none of it shows on the surface. The owner sees a "working" site; visitors bounce, screen-reader users give up, Google ranks it down.

Expensive

A professional audit costs \$300–\$1000, takes days, and arrives as a PDF of jargon. For 200M+ small-business sites, that math never works — so audits simply don't happen.

Unactionable

Free checkers scan one page, print generic advice ("optimize your images"), hide their failures, and never tell a developer WHICH element to change or WHAT code to paste.

The gap: nothing between a \$0 toy and a \$1000 consultant — no tool that finds real problems, proves them with evidence, and hands over the exact fix.

Every pain point, answered directly.

😞 "Audits cost \$1000 and take days"

Out of reach for the businesses that need them most.



Paste a URL. Minutes later: full diagnosis.

Real crawl of up to 10 pages, 60+ checks, Lighthouse, axe-core, security + TLS — automatically, for the cost of a coffee's worth of compute.

😞 "Reports tell me WHAT's wrong, never HOW to fix it"

Findings without fixes just create anxiety.



Every issue ships its fix — with the broken element attached.

Business impact in plain language, AI-written paste-in code, and the actual HTML of the offending element with its selector. Search, paste, done.

😞 "Free checkers can't be trusted"

Silent failures dressed up as green ticks.



Honesty is enforced by the database.

Every failed check = exactly one issue, counts reconcile everywhere; anything that couldn't run says so in orange with a one-click re-run. Never a fake pass.

From URL to fixed website — six steps.

1 Paste a URL

One input. No setup, no snippet to install, no account gymnastics.

2 Real crawl

Headless Chromium visits up to 10 pages — real navigation, screenshots, full HTML, response headers. Live progress streams to the screen.

3 Deep audits

60+ rule checks + Lighthouse (LCP/CLS/TBT) + axe-core accessibility + security headers, TLS cert, broken links — per page.

4 AI review

The model reviews each screenshot like a senior designer, audits the markup, checks cross-page consistency, and writes an executive summary.

5 Issues → fixes

Every failed check becomes one prioritized issue: impact, paste-in code, and the exact broken elements captured from the page.

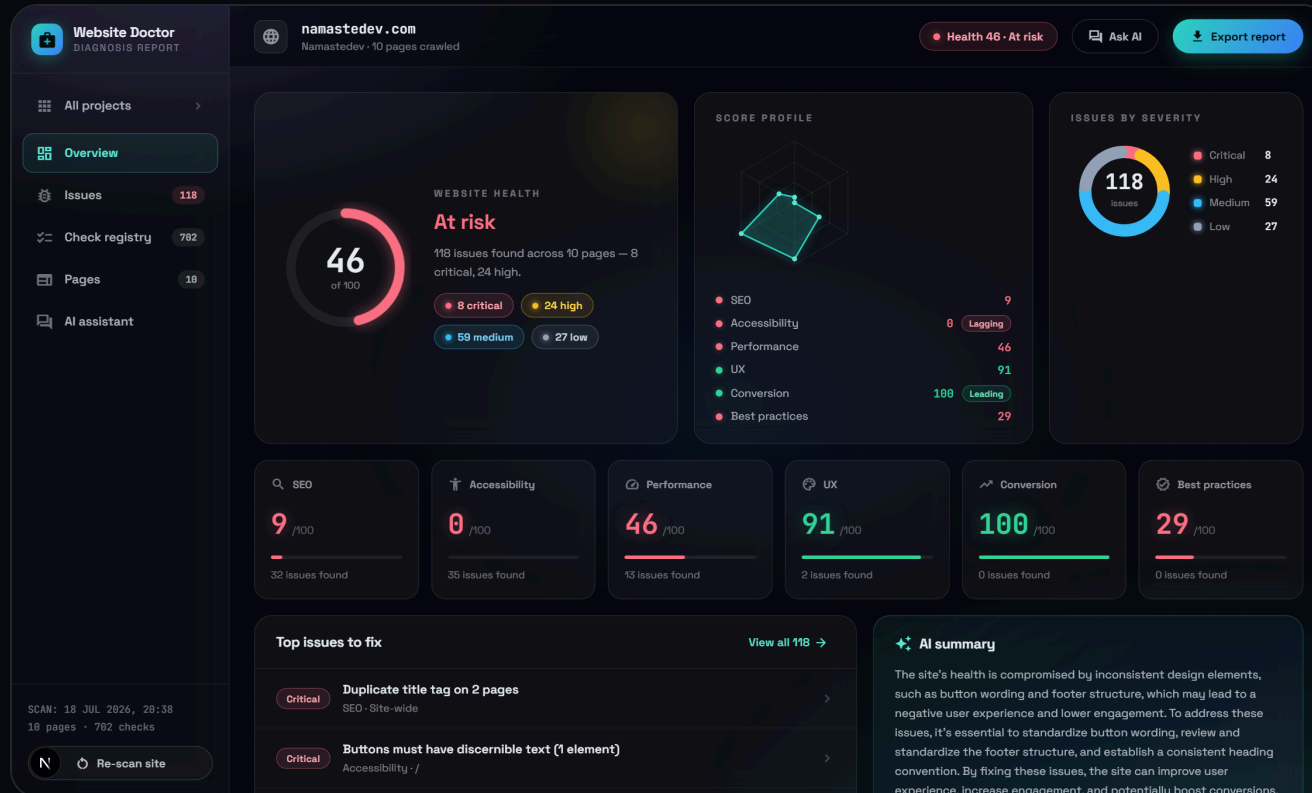
6 Act & verify

Mark resolved, re-run single checks to confirm fixes, ask the AI assistant anything about the findings, export the PDF for the client.

`crawl` → `audit` → `AI` → `issues` → `fixes` → `verify` — a full quality loop, not a one-shot report.

THE PRODUCT

One URL in. A complete diagnosis out.



- ✓ Health score + 6 category scores, severity-weighted
- ✓ Issues with evidence, impact, and copy-paste fix code
- ✓ The actual broken HTML elements, captured from the page
- ✓ Live 15-stage scan progress with real telemetry
- ✓ Check registry: 393 checks, honest statuses, 1-click re-run
- ✓ PDF export + site-aware AI chat

AI is the engine, not a wrapper.

Vision review

Reviews every page's real screenshot like a senior designer — CTA contrast, hierarchy, trust signals — findings join the issue pipeline with full 1:1 rigor.

Markup review

Reads real extracted DOM data and catches what static rules can't: weak titles, mismatched schema, suspicious anchors — grounded, never invented.

Fix generator

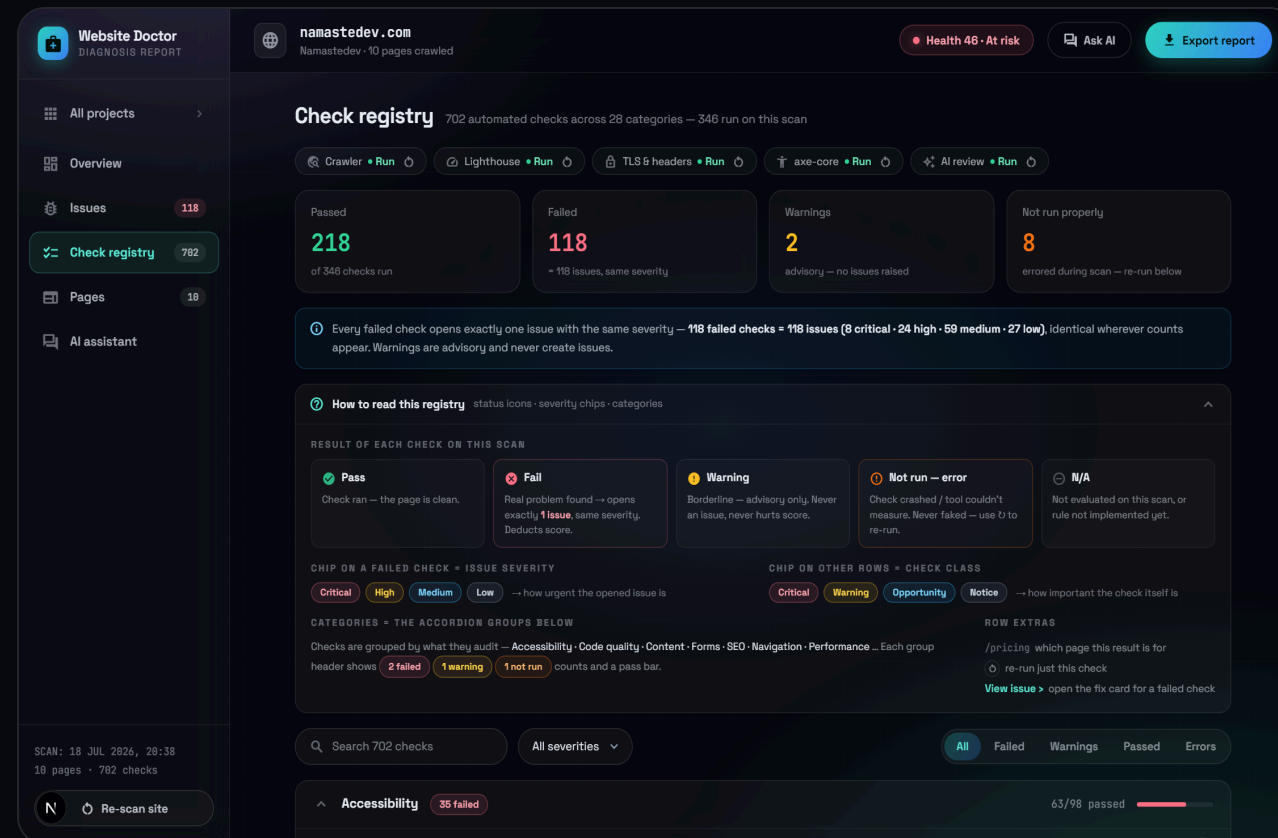
Writes each fix card from the actual offending element's HTML — page-specific paste-in code, not generic advice.

Grounded chat + resilience

Chat answers strictly from stored analysis and refuses the rest. Provider-agnostic (Groq/Gemini/any OpenAI-compatible), zod-validated JSON, and **self-healing model fallback** when free-tier quotas run dry.

WHY IT'S DIFFERENT

The honest scanner.



✓ **1:1 invariant, DB-enforced** — failed checks = issues, same severity, counts reconcile in every view

✓ **No silent failures** — "Not run — error" rows with the real reason. Never a fake pass, never a fake 0/100

✓ **One-click re-run** — per check or per analyzer, offline from stored artifacts where possible

✓ **Resolved marks survive re-runs** — the tool respects your triage

Trust is the product. If users can't trust the numbers, an audit tool is decoration.

Everything on this slide is live right now.

- ✓ **Real crawl** — up to 10 pages, headless Chromium: screenshots, HTML, headers, live SSE progress + telemetry
- ✓ **62 running checks** — 27 rules + Lighthouse (LCP/CLS/TBT) + axe-core + security headers/TLS + broken links (full 331-check catalog registered, honestly labeled N/A until implemented)
- ✓ **Full AI layer** — vision review of real screenshots, markup review, cross-page executive summary, AI fix cards, grounded chat
- ✓ **Affected elements** — every issue carries the real broken HTML + selector, captured from the page
- ✓ **Honesty engine** — DB-enforced 1:1 invariant, "Not run — error" states, per-check & per-analyzer re-run, resolved marks survive re-runs
- ✓ **Self-healing ops** — automatic model fallback on quota exhaustion, rate-limit hopping, stuck-job watchdog, boot self-seeding on fresh deploys
- ✓ **Provider-agnostic AI** — Groq / Gemini / any OpenAI-compatible endpoint, zod-validated JSON, swap via env only
- ✓ **Ship-ready** — PDF report export, multi-site projects, responsive UI, 33 unit tests + a live end-to-end smoke gate on every change

Deployed and clickable: live demo self-seeds two scanned sites — judges can open reports, re-run checks, and chat with the analysis immediately.

Capped on free tiers. Uncapped in one env change.

- ✓ **100% free infrastructure** — free-tier server + Groq free AI. Demo caps exist only to respect their rate limits: vision on 2 pages, Lighthouse on 5, 3s between AI calls
- ✓ **The engine has no limits** — clone, pnpm dev, and the same pipeline audits all 10 pages at full depth: every analyzer, every AI review
- ✓ **Rate-limited ≠ hidden** — a throttled stage shows an honest **"Not run — error"** row with the real reason and a one-click re-run. Results are never faked

We treat rate limits like failed checks: transparently. A QC tool that fakes its own results can't be trusted to QC anything.

```
# .env — full-potential local run
```

```
LIGHTHOUSE_MAX_PAGES=10
```

```
AI_MAX_VISION_PAGES=10
```

```
AI_MAX_HTML_PAGES=10
```

```
AI_MAX_FIX_ISSUES=50
```

```
AI_CALL_GAP_MS=0
```

One file. Zero code changes. Full potential.

Every website is a customer.

200M+

active sites; wedge = agencies turning
"trust us" into branded, evidence-backed
reports with fixes

Freemium

free scans → paid monitoring, multi-site
dashboards, white-label PDFs

Next

scheduled re-scans + regression alerts ·
auto-fix pull requests · CI gate that fails
deploys when health regresses



Website Doctor

live demo · github.com/kmr22live/website-doctor · built with AI in 4 days